

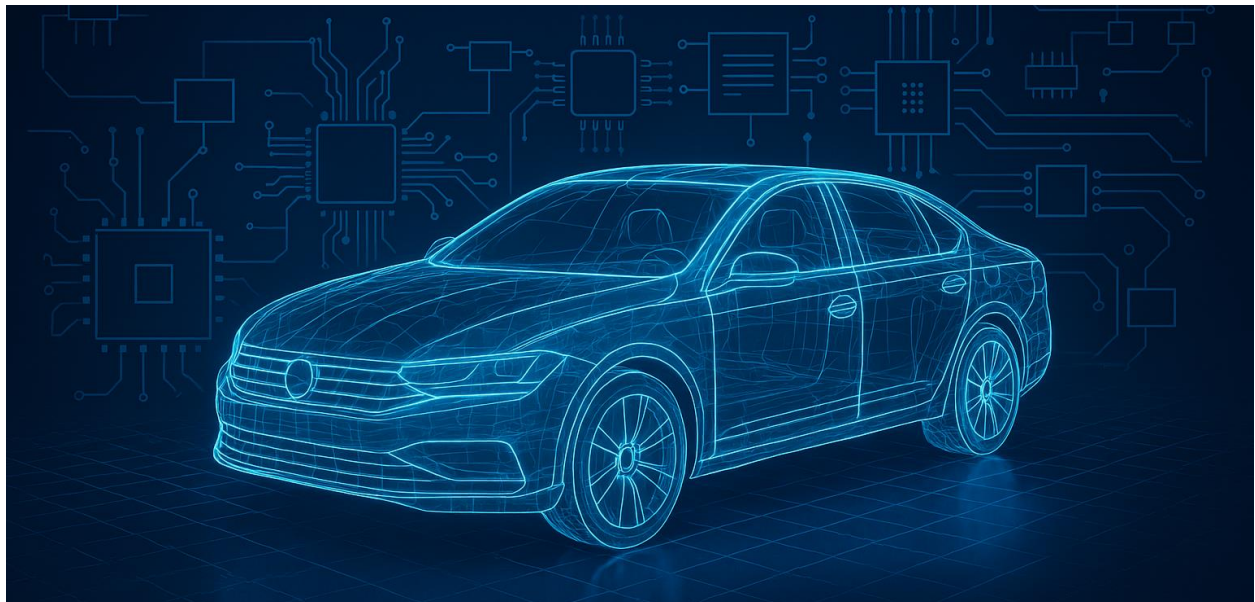


Architecting the Future: Technical Foundations of Software-Defined Vehicles

Introduction

The shift from traditional hardware-centric vehicles to software-defined vehicles (SDVs) marks the industry's embrace of digital transformation, where software becomes the core enabler of innovation, functionality, and business models. In this paper, we examine the challenges and opportunities of SDVs, the architectural changes they necessitate, and how these new platforms will utilize collaborative ecosystems.

A 'software-defined vehicle (SDV)' is an automobile that relies on digital technology. All functionalities, including driving, entertainment, communication, safety, and comfort, are enabled, managed, controlled, and customized through software.¹



¹ <https://www.marketsandmarkets.com/Market-Reports/software-defined-vehicles-market-187205966.html#:~:text=What%20is%20the%20current%20size,at%20a%20CAGR%20of%2034.0%25>

The Shift to Software-Defined Vehicles

A software-defined vehicle is one whose core functions, including steering, braking, infotainment, cabin control, battery management, and more, are controlled and enhanced through software. This transformation enables the development of more advanced driver assistance systems (ADAS), autopilot features, and even AI-driven personalization, such as selecting a restaurant based on user preferences.

The modern vehicle dashboard has turned into a “Digital Cockpit” and increasingly resembles a tablet, complete with apps and a familiar operating system, and where mechanical actuators are replaced by softkeys. These apps can add new features and customizations, ranging from internal light coordination and weather app integration to heated or cooled seats, and possibly even game console compatibility. Features such as steering behavior, once mechanically defined, are now programmable, offering customizable driving experiences. This shift mirrors the evolution of mobile phones, where hardware and software were once tightly coupled but are now part of a flexible, app-driven ecosystem.

Traditional vehicles rely on a complex network of electronic control units (ECU), each responsible for specific tasks, such as parking sensors or dashboard alerts. These systems are often fragmented, each with its own microcontroller, firmware, and software written in different languages by various suppliers, making integration and updates cumbersome. In contrast, SDVs consolidate computing power into centralized computing and zonal architectures, significantly reducing the number of ECUs, simplifying the vehicle's internal structure, and enhancing connectivity with cloud services. This decoupling of software and hardware enables layered architectures that support safer, more intelligent, and upgradable vehicles.

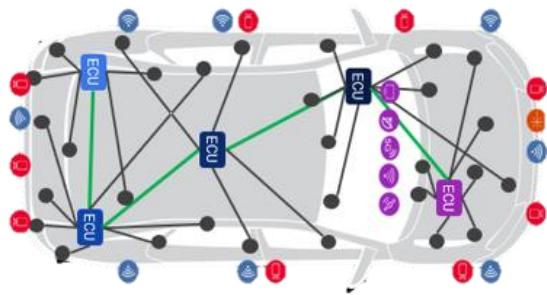


Figure 1. A traditional vehicle with a domain-oriented architecture, where software is embedded in a distributed hardware architecture, and domains are connected across the vehicle, is heavy, costly, and static.

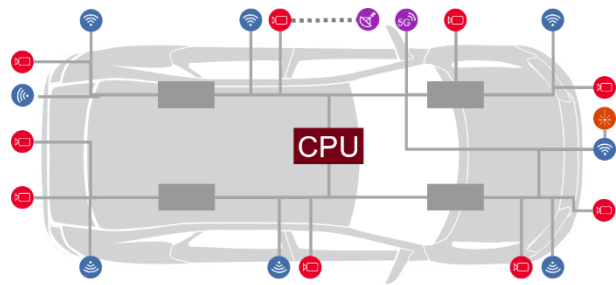


Figure 2. SDV with a centralized computer and zonal architecture seeks to consolidate decision-making, leaving edge actuators controlled by commands from the central computer to optimize power, save weight, reduce complexity, and lower costs.

The shift from domain-oriented to service-oriented architecture is a key aspect of SDV development. Traditional vehicles may contain over 50 kg of copper wiring due to numerous ECUs and communication buses. SDVs, however, consolidate functionality into four zones (based on location: front right, front left, back right, and back left) with a central high-performance computer, significantly reducing wiring and weight. This not only simplifies design and maintenance but also improves vehicle efficiency and performance.

Service-oriented architecture is a software design approach that focuses on creating reusable services that work independently but also communicate with one another and can be upgraded or replaced without re-engineering the entire system.

This new architecture supports high-bandwidth data transfer, which is essential for processing inputs from multiple sensors, such as radar, lidar, and cameras. It also allows for over-the-air (OTA) updates, enabling continuous software improvements and new feature rollouts without requiring physical service visits. This evolution enables manufacturers to continually improve vehicle performance, safety, and user experience through software enhancements, without requiring major hardware changes. By leveraging data analytics and connected sensors, these vehicles can adapt to changing conditions and customer needs in real time.

New Business Models and Ecosystems

SDVs are not just a technological shift; they represent a new business paradigm. Automakers can now offer features as subscription services, like those found in app stores on smartphones. This opens recurring revenue streams and enables customers to personalize their vehicles after purchase. Tesla has already started offering these monthly subscriptions.

To support this, OEMs must build robust ecosystems that include cloud platforms, cybersecurity workflows, and developer-friendly APIs. Hyundai's Pleos platform exemplifies this approach, enabling third-party developers to create applications for vehicles, much like mobile app ecosystems. This strategy reduces the need for OEMs to develop all software in-house and fosters innovation through external contributions.

The simplified architecture of SDVs reduces hardware complexity and increases reliance on software, which in turn opens opportunities for new players in the automotive ecosystem, including platform and software providers, alongside traditional chipset vendors and Tier 1 suppliers.

Evolution of SDV Capabilities

According to PricewaterhouseCoopers, SDVs evolve through levels 0 to 5, similar to the SAE levels of vehicle autonomy.²

Level 0 is a mechanically controlled vehicle, equipped with micro-controllers and firmware-based functionality

Level 1 is an Electrical and Electronic (E/E)-controlled vehicle with discrete Electronic Control Units for each function. Software is introduced for basic driver assistance, enabling isolated features like lane keeping and cruise control.

Level 2, OTA functionality is available, limited to infotainment-related updates and software bug fixes, ie, for recalls, navigation, and security updates.

Level 3 shifts to domain architecture and offers advanced ECU integration with active software updates via OTA, not only to address issues but also to implement functional updates.

Level 4 is a fully software-defined vehicle that optimizes functions through frequent and proactive OTA updates, adding new features and enhancing product value. Revenue models beyond vehicle sales are established.

Level 5 is not only a software-defined vehicle but also a software-defined ecosystem. Seamlessly and always connected, to enable software updates within the vehicle, and also to continuously collect data from in and outside the vehicle.

As levels increase, software and services become more central, while the complexity of the electronic architecture decreases. Levels 3 and above are defined as SDV. The transition to zonal architecture occurs at Level 4. The goal is to ensure user needs are constantly met. In the realm of infotainment, common apps and services can be provided regardless of the vehicle, significantly shifting the value of mobility towards software and services. These advancements elevate the overall value of mobility across the ecosystem, continuously maximizing and updating user value and experience.

The most advanced companies like Tesla, Rivian, NIO, and XPENG Motors, as recently named by Motor Trend ³ are operating at or near a Level 4, with most others aiming to transition to Level 3, depending on their technological maturity.

² <https://www.pwc.com/jp/en/knowledge/column/definition-of-sdv.html>

³ MotorTrend Announces Winners of the 2025 Software-Defined Vehicle Innovator Awards

To shorten the cycle, development and test processes will need to be automated, and more simulation modeling and digital twins will need to be utilized, gradually incorporating hardware to bridge the gap between simulation and road testing. With collaboration platforms for continuous testing accessible in the cloud for real-time learning and adjustments.

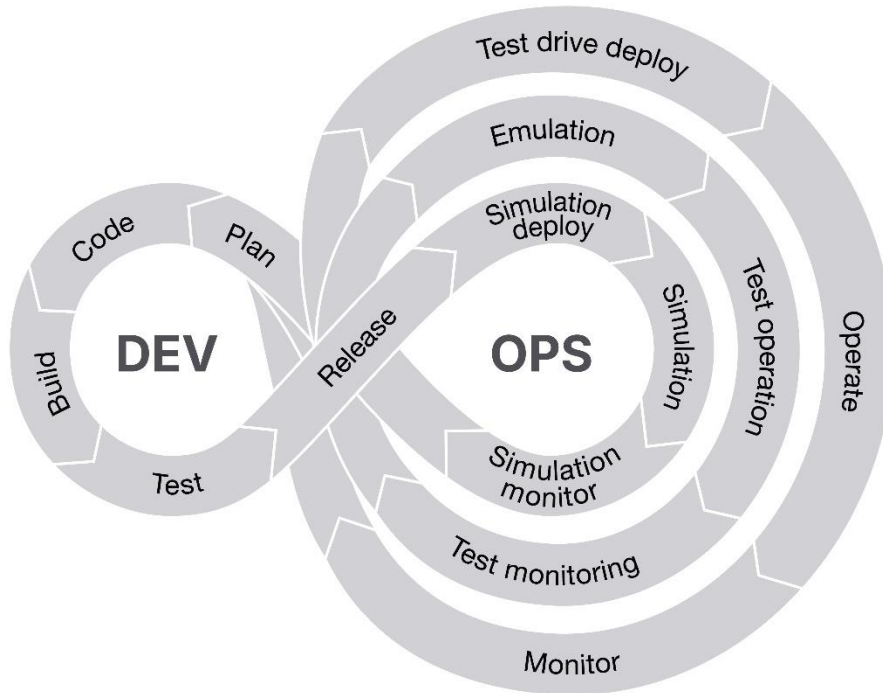


Figure 5. Automakers build on prior releases first with simulation, then incorporate a little hardware into the software loop, and finally, learn from the vehicle's real-world driving. Each learning is integrated into the model to accelerate development time.

The result from the simulation feeds into emulation and then into road testing. Each loop builds on the previous one. Design and test engineers employ this test process by replacing fundamental components of the simulation, such as a brake or a steering wheel subsystem.

Another key challenge is ensuring safety and cybersecurity, as vehicles become increasingly connected and software-based, they must be protected against hacking and software vulnerabilities. Cybersecurity becomes increasingly complex with the addition of each new interface to the vehicle. There are also additional requirements for regulations to ensure robust security protocols that protect both the vehicle and the user.

To address these, OEMs are adopting DevSecOps models, which integrate development, security, and operations into a continuous cycle. This approach aims to reduce the risk of releasing code with security vulnerabilities by making security a shared responsibility among development, security, and operations teams. With fewer, more centralized control units, manufacturers can better manage potential vulnerabilities. Strategies such as DevSecOps and zero-trust frameworks ensure that only verified communications and updates are accepted by the vehicle.

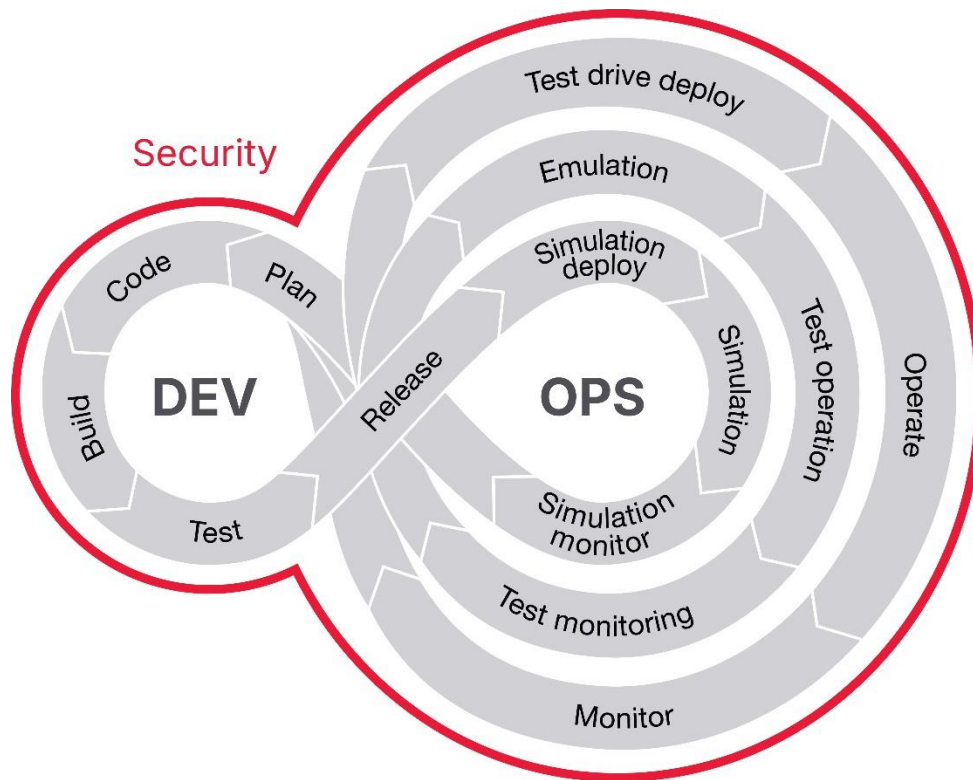


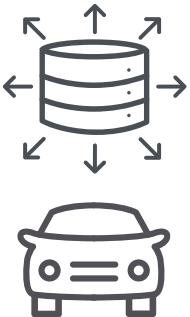
Figure 6. DevSecOps model adds an integrated layer of security to the continuous cycle of development, test, release, and refresh.

Another challenge that OEMs are perhaps the most eager to face **is enabling new business models**. They can seamlessly integrate subscription services and OTA updates to monetize new features and functionalities. The shift towards software-defined vehicles drives the need for solutions that enhance user experience and operational efficiency. This includes the integration of advanced telematics, real-time data analytics, and connectivity features that support seamless communication between vehicles, infrastructure, and users. By leveraging these technologies, OEMs can offer personalized services, optimize fleet management, and ensure continuous business operations. Data analytics plays a crucial role in identifying user needs and potential services, while modular architecture allows for scalable and flexible vehicle platforms.

Enabling the SDV Ecosystem

There are several new processes and technologies that will serve as catalysts and empower this SDV transformation. These four are crucial for addressing the challenges.

Process and Technology enablers

	Scalable and Modular Architecture	Vehicles are built with hardware that can support future software updates, much like smartphones. This means new features can be added over time, often by third-party developers. Flexible hardware and software platforms that support the integration of third-party applications and enable the rapid innovation and development of new features.
	Data Analytics	OEMs must collect, analyze, and leverage the vast amounts of data generated by vehicles to enhance performance, create new services and revenue streams, such as predictive maintenance, subscription-based services, or feature unlocks.
	Cybersecurity	Protecting vehicle systems from malicious actors is critical. This includes securing infotainment systems, communication channels, and ensuring that OTA updates are authentic. Ensuring robust security protocols to protect the vehicle and user data from breaches, hacking, or malicious attacks.
	Over-the-Air Updates	Frequent and seamless software updates are expected by consumers and are essential for maintaining functionality and security. The ability to deploy software updates, patches, and new features remotely to enhance functionality and improve user experience without requiring physical intervention.

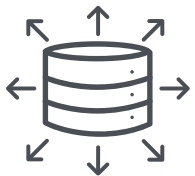
Scalable and Modular Architecture



SDVs impact the entire supply chain, not just OEMs. Tier One and Tier Two suppliers will need to work together to support the automotive OEMs. Unlike mechanical parts, software from different suppliers may interact with each other as data passes between devices. When problems occur, multiple parties are required to replicate the symptoms to troubleshoot the issues.

Some OEMs will continue to form partnerships with technology companies and suppliers to accelerate their SDV development. For example, Valeo is working to act as a system integrator for SDV solutions, leveraging its software expertise. All OEMs are working on expanding and improving their existing software platforms. This includes integrating new features and enhancing interoperability between different systems. For example, the Eclipse consortium with a Software Defined Vehicle "S-CORE", providing an open-source-based SDV platform.

Data Analytics



The increased functionality and customization offered by SDVs enable greater personalization of features and services, tailored to individual user preferences. This personalization can lead to increased customer satisfaction and loyalty, ultimately driving revenue growth. Most OEMs (81%) expect to generate significant revenue from data monetization, indicating broad confidence in the financial potential of data-driven services. The average expected revenue is around US\$720 million over the next five years ⁴.

Data analytics will be involved in collecting and analyzing vast amounts of vehicle and user information to improve performance, predict maintenance needs, and create new revenue streams. This data will then be used to offer personalized services, optimize vehicle operations, and develop subscription-based models for features and services.

For example, Tesla uses data analytics to monitor the health of its vehicles in real time. By analyzing data from various sensors, Tesla can predict when a component is likely to fail and notify the owner to schedule maintenance before a breakdown occurs. This not only enhances vehicle reliability but also reduces maintenance costs and downtime. Another option is subscription-based services where drivers can subscribe to specific software features or apps, introducing a consistent revenue source for OEMs. Imagine integrating Amazon Alexa into your vehicle, allowing you to purchase items you've been waiting to buy at a traffic light.

⁴ Deloitte 2024 Global SDV Readiness Survey.

Enhanced Cybersecurity Measures



Cybersecurity has emerged as one of the most critical concerns for automotive manufacturers. With each new interface added to the vehicle, risk assessment also increases. In addition to protecting users and their own data, regulatory requirements such as UNECE R155 and R156.6 are now in place. Car manufacturers that want to sell a vehicle in Europe must comply with European regulations. The complexity of achieving comprehensive cybersecurity has increased, making it difficult even for large OEMs to keep up with evolving threats.

"Partnerships in cybersecurity are important for OEMs to leverage external expertise, meet evolving regulatory requirements, and manage risk. By sharing resources and intelligence, OEMs can transform cybersecurity into a strategic advantage."

Ingo Dassow, partner, Deloitte Consulting GmbH

Definitions

UNECE R155 and R156.6: United Nations Economic Commission for Europe (UNECE) regulations addressing cybersecurity management systems (CSMS) and Software update management systems (SUMS). These regulations aim to standardize practices across manufacturers, suppliers, and software providers to secure the automotive system and ecosystem.

ZeroTrust: Continuous Authentication, Least Privilege Access, Data Encryption, Micro Segmentation & Cyber Isolation

SecOps: Incident Response Automation, Compliance Management & Badging, Security Awareness & Training

Enable Seamless Over-the-Air Updates



SDVs will receive continual OTA software updates that will close security gaps quickly and implement new security functions immediately. Advancing autonomous driving functions and ensuring safety are key priorities, but one of the ways the cost benefits can be immediately recognized is as a solution to address vehicle recalls, through the implementation of OTA updates. These updates allow manufacturers to remotely fix software issues without requiring the vehicle to be taken to a service facility. This capability helps to enhance the user experience by minimizing disruptions and provides significant advantages in recall avoidance and cost savings. In June 2024, an OEM “recalled” 1.85 million vehicles in the United States due to software that failed to detect improperly closed hoods, increasing the risk of accidents. Instead of physically recalling affected vehicles to service facilities, the OEM deployed an OTA update that modified the software to recognize an open hood and promptly alert the driver. This approach clearly demonstrates the effectiveness of OTA updates to swiftly address obvious safety concerns in a cost-efficient manner.

Summary

Software-defined vehicles represent a fundamental shift in how cars are designed, built, and experienced. By centralizing computing, enabling OTA updates, and fostering open ecosystems, SDVs offer unprecedented flexibility and innovation. As the industry evolves, collaboration, cybersecurity, and customer trust will be key to unlocking the full potential of SDVs and shaping the future of mobility.

The transition to software-defined vehicles marks a pivotal moment in automotive history. It brings with it a host of opportunities—from enhanced safety and convenience to entirely new business models. However, it also introduces significant challenges, particularly in cybersecurity and system integration.

Read more about Keysight’s role in enabling SDVs at [blog 4](#).

Keysight enables innovators to push the boundaries of engineering by quickly solving design, emulation, and test challenges to create the best product experiences. Start your innovation journey at www.keysight.com.



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